

MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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Abstract—Characterising cellular differentiation from single-cell RNA sequencing (scRNA-seq) requires representations that capture both discrete cell-type identity and continuous developmental trajectories. We present MoCoO, a modular framework integrating a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo), complemented by an optional Phase-2 Flow Matching (FM) refinement step. Through a systematic six-configuration ablation across 20 scRNA-seq datasets evaluated with a proposed six-metric suite covering clustering geometry (ASW, DAV) and embedding quality (DRE, LSE, DREX, LSEX), we demonstrate two central findings. First, the ODE+MoCo combination is the core architectural synergy: VAE+ODE+MoCo achieves four of six top-two finishes among base configurations, including the best ASW (0.225) and DAV (1.478), plus second-best DRE (0.640) and LSE (0.335). Second, FM refinement systematically improves latent-space embedding quality—DREX in 92% and DRE in 88% of 120 dataset-configuration pairs ($\Delta\text{DREX} = +0.030$, $\Delta\text{DRE} = +0.023$). Combined, VAE+ODE+MoCo+FM ranks top-two on five of six metrics across all 12 configurations. ODE smooths the latent manifold along developmental trajectories; MoCo sharpens cluster geometry; FM recovers and amplifies embedding quality post-hoc. Downstream validation confirms that MoCoO latent spaces support annotation transfer, uncertainty quantification, differential expression, and branching detection. Pseudotime predictions correlate significantly with canonical marker genes across all five core developmental systems. We publicly release the MoCoO Python package (`pip install mocoo`) and full benchmark suite.

Index Terms—Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology.

I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [1]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type identity and *continuous* developmental dynamics such as differentiation trajectories [2], [3]. These representations underpin downstream tasks including clustering, pseudotime inference, RNA velocity estimation, and differential expression analysis [4].

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Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [5] and scVAE [6] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend to be over-smoothed, conflating nearby cell states. Two complementary strategies address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [7] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the latent derivative with a neural network and integrating via an ODE solver, these models learn smooth trajectories without discrete time-step assumptions. Latent ODE models [8] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [9].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [10]—learns representations that are locally smooth yet globally discriminative. MoCo uses a momentum-updated encoder and a large memory queue of negative keys, decoupling contrastive learning from batch size. Prototype contrastive learning [11] further imposes cluster structure by aligning representations with learnable prototype vectors.

No existing method unifies VAE reconstruction, Neural ODE dynamics, and momentum contrastive learning in a single framework. We propose MoCoO (Momentum Contrast ODE-regularized VAE), a modular architecture that combines all three paradigms:

- 1) **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space that handles zero-inflation and overdispersion in scRNA-seq counts.
- 2) **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
- 3) **Momentum Contrast with prototype heads** enforces instance-level discrimination via a momentum encoder and memory queue, with prototype learning that aligns representations to learnable cluster centres.
- 4) **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.

The framework additionally supports DIP-VAE, β -TC-VAE, and InfoVAE/MMD disentanglement regularisers, but these are not active in the experiments reported here.

We conduct a systematic ablation across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and VAE+ODE+MoCo+Proto) on 20 scRNA-seq datasets spanning diverse developmental systems, with an optional Phase-2 Flow Matching (FM) refinement step that extends this to 12 configurations. Deep-learning external baselines with comparable latent representations—scVI [5], Harmony [12], and scANVI [13]—contextualise MoCoO’s absolute performance. A key finding is that combining ODE and MoCo consistently produces the best cluster geometry (cross-dataset mean ASW 0.225, DAV 1.478 for VAE+ODE+MoCo), while VAE+ODE achieves the best embedding quality (DRE 0.647, DREX 0.613). ODE-containing configurations consistently outperform non-ODE variants on cluster geometry across all 20 datasets ($\Delta\text{ASW} \sim 0.04$, $\Delta\text{DAV} \sim 0.23$). FM refinement improves distance-preservation metrics in over 85% of dataset–configuration pairs (DREX $\Delta = +0.030$, DRE $\Delta = +0.023$). Component analysis reveals that each module contributes complementary geometric structure: the Neural ODE smooths the latent manifold along trajectories and MoCo sharpens pairwise distance relationships, with the ODE+MoCo combination producing the strongest overall synergy across all 20 datasets. Evaluation is centred on a proposed six-metric suite covering clustering geometry (ASW, Davies–Bouldin) and embedding/latent-space quality (DRE, DREX, LSE, LSEX), complemented by biological validation (pseudotime–marker correlations), downstream analysis (annotation transfer, uncertainty quantification, generation quality, gene importance, branching detection, differential expression), batch integration (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB [14]), and training dynamics.

II. RELATED WORK

A. Variational Autoencoders for scRNA-seq

scVI [5] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [6] (multiple count distributions), scANVI [13] (semi-supervised cell-type annotation), and totalVI [15] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

B. Neural ODEs and Latent ODE Models

Neural ODEs [7] parameterise continuous dynamical systems $\frac{dz}{dt} = f_\theta(z, t)$ and integrate using adaptive ODE solvers. The Latent ODE framework [8] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [9], while related dynamical modelling approaches have advanced RNA velocity estimation [16]. Optimal-transport-based methods such as TrajectoryNet [17] and CellOT [18] model population-level dynamics rather than single-cell ODE trajectories. MoCoO extends the PanODE framework [9] (by the first author). Specifically, MoCoO inherits the VAE encoder–decoder, Neural ODE integration, and pseudotime-prediction head from PanODE. The novel contributions of the current work are: (i) the MoCo module

with momentum encoder and memory queue, (ii) learnable prototype heads for cluster anchoring, (iii) the information bottleneck, (iv) the systematic six-configuration ablation design, and (v) the DRE/DREX/LSE/LSEX evaluation suites. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

C. Contrastive Learning for Single-Cell Data

MoCo [10] and SimCLR [19] have been adapted to biological contexts. scAGCL [20] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [21] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

D. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [22] uses diffusion models (related to score-based SDEs) for single-cell generation. VeloVAE [23] integrates VAE with RNA velocity. Diffusion pseudotime (DPT) [24] infers trajectories through diffusion maps of the k -NN graph, while Monocle 3 [25] and PAGA [26] use graph abstractions for trajectory inference, Slingshot [27] fits simultaneous principal curves for lineage-specific pseudotime, and Palantir [28] models cell-fate probabilities via diffusion maps for multiscale pseudotime estimation. For batch integration, Harmony [12] applies iterative soft clustering in PCA space to remove batch effects. However, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

III. METHOD

A. Overview

MoCoO (Fig. 1) processes scRNA-seq count data $X \in \mathbb{R}^{N \times G}$ (N cells, G genes) through five components: (1) a VAE with count-based decoder, (2) a Neural ODE for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

B. Encoder

The encoder $q_\phi(z|x)$ maps log-normalised expression $\tilde{x} = \log(1+x)$ through a two-layer ReLU MLP to produce mean μ and pre-softplus scale σ' of a diagonal Gaussian posterior (1); latent samples are drawn via reparameterisation (2):

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(\tilde{x}), \text{softplus}(\sigma'_\phi(\tilde{x}))^2 I), \quad (1)$$

$$z = \mu + \text{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

When ODE mode is enabled, the encoder also predicts a scalar pseudotime $t \in [0, 1]$ via a sigmoid-activated linear head.

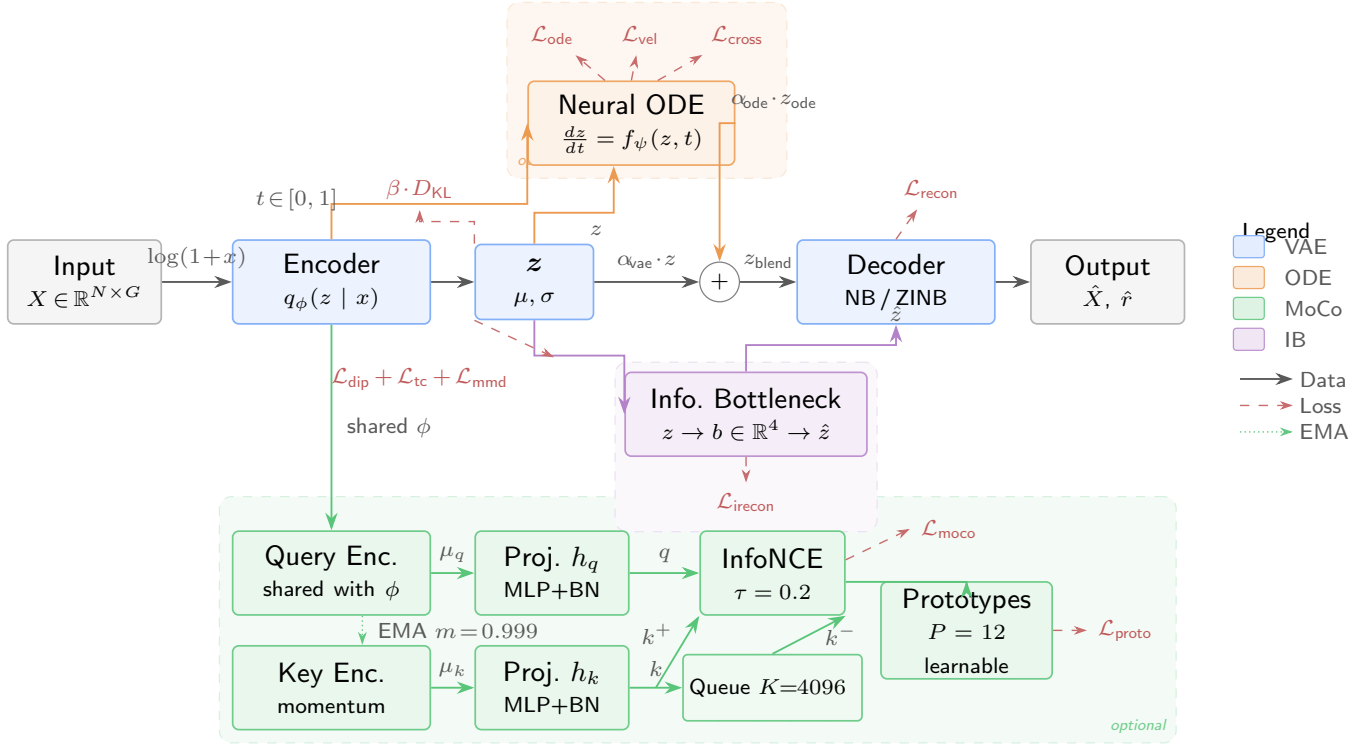


Fig. 1. MOCoO architecture overview. Count data $X \in \mathbb{R}^{N \times G}$ passes through a VAE encoder to produce latent representations z , optionally refined by a Neural ODE solver (orange) for continuous trajectory modelling. Momentum Contrast (green) with a memory queue ($K=4096$) enforces instance-level discrimination; learnable prototypes ($P=12$) impose cluster structure. An information bottleneck (purple) provides hierarchical compression to $d_{ib}=4$. Dashed regions denote optional modules. Colour coding is consistent throughout all figures.

C. Neural ODE Component

Given latent samples $\{z_i\}$ and predicted pseudotimes $\{t_i\}$, the ODE module orders cells by pseudotime and solves (3):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\text{init}}, \quad (3)$$

where f_ψ is a two-layer ELU MLP. Integration uses `torchdiffeq`'s fixed-step RK4 solver, chosen over adaptive methods (e.g., Dormand–Prince [29]) because the encoder-predicted pseudotime grid is relatively smooth and fixed-step integration provides deterministic gradients without adaptive step-size overhead—important for stable joint optimisation of the encoder and ODE function. Preliminary comparisons showed negligible clustering geometry differences ($\Delta\text{ASW} < 0.005$, $\Delta\text{DAV} < 0.02$) between RK4 and Dormand–Prince adaptive integration, with $\sim 30\%$ longer wall-clock time for the adaptive solver due to step-size adaptation overhead in the joint training loop. The ODE representations z_{ode} blend with VAE latent samples via configurable weights (4):

$$z_{\text{blend}} = \alpha_{\text{vae}} \cdot z + \alpha_{\text{ode}} \cdot z_{\text{ode}}. \quad (4)$$

Two auxiliary losses regularise the ODE:

- **ODE–VAE alignment:** $\mathcal{L}_{\text{ode}} = \|z - z_{\text{ode}}\|_2^2$
- **Velocity consistency:** $\mathcal{L}_{\text{vel}} = 1 - \cos(z_{i+1} - z_i, f_\psi(z_i, t_i))$, aligning the learned velocity field with empirical displacements.

D. Momentum Contrast Module

The MoCo module follows MoCo v1 [10] with enhancements from scAGCL [20] and scGPCL [21]:

- 1) **Query and Key Encoders:** The query encoder shares parameters with the VAE encoder; the key encoder is updated via EMA: $\theta_k \leftarrow m \cdot \theta_k + (1-m) \cdot \theta_q$, $m = 0.999$.
- 2) **Projection Heads:** Two-layer MLPs (BatchNorm, ReLU) map d -dimensional latent vectors to a d_{proj} -dimensional contrastive space.
- 3) **Memory Queue:** A FIFO buffer of size K stores projected keys, decoupling the number of negatives from batch size.
- 4) **InfoNCE Loss** (see (5)):

$$\mathcal{L}_{\text{moco}} = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{k^-} \exp(q \cdot k^- / \tau)}, \quad (5)$$

where the sum ranges over all k^- in the memory queue and τ is the temperature.

- 5) **Prototype Contrastive Loss** (optional; see (6)):

$$\mathcal{L}_{\text{proto}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(z_i \cdot p_{c(i)} / \tau)}{\sum_{j=1}^P \exp(z_i \cdot p_j / \tau)}, \quad (6)$$

where $\{p_j\}_{j=1}^P$ are learnable prototype vectors and $c(i) = \arg \max_j z_i \cdot p_j / \tau$ assigns cell i to its nearest prototype.

Positive pair construction. For each mini-batch, two stochastic augmented views of the same cell form positive

pairs: the query view passes through the main encoder, while the key view passes through the momentum encoder. The augmentation pipeline applies three transformations, each with probability $p_{\text{aug}}=0.5$: (1) gene masking—randomly zeroing out genes with probability 0.1; (2) Gaussian noise—adding $\mathcal{N}(0, 0.04)$ to randomly selected genes (probability 0.1); (3) feature swapping—exchanging gene values across cells in the batch (probability 0.05). Negatives come from all keys stored in the memory queue.

Augmentation rationale: Gene masking simulates dropout, which is a dominant noise source in scRNA-seq protocols [30]. Gaussian noise ($\sigma=0.2$) models technical variability in library preparation. Feature swapping operates at very low probability ($p=0.05$, affecting $\sim 5\%$ of genes per cell), which is equivalent to a regularised mixup—it encourages invariance to minor expression perturbations without creating biologically implausible hybrid profiles. All three augmentations are deliberately conservative: masking affects only 10% of genes per view, noise is applied independently per gene, and swapping probability is below the natural inter-cell variance within a cell type. This design follows the principles validated in scAGCL [20] and scGPCL [21], which demonstrated that moderate stochastic augmentations improve contrastive learning for scRNA-seq without distorting cell-state semantics.

E. Decoder

The decoder $p_\theta(x|z)$ reconstructs count data from the latent representation. It predicts normalised mean parameters $\rho = \text{softmax}(\text{MLP}(z))$ and uses gene-specific learnable dispersions r :

- **Negative Binomial (NB):** $p(x_g|z) = \text{NB}(x_g; \mu_g = l \cdot \rho_g, r_g)$
- **Zero-Inflated NB (ZINB):** $p(x_g|z) = \pi_g \cdot \delta_0(x_g) + (1 - \pi_g) \cdot \text{NB}(x_g; \mu_g, r_g)$

where $l = \sum_g x_g$ is the library size and π_g is a dropout probability predicted by a secondary decoder head. While MoCoO supports MSE, NB, ZINB, Poisson, and ZIP likelihoods, all experiments in this paper use the negative binomial (NB) likelihood unless otherwise stated.

F. Information Bottleneck

A secondary encoder-decoder pair maps $z \in \mathbb{R}^d$ to a bottleneck $b \in \mathbb{R}^{d_{\text{ib}}}$ ($d_{\text{ib}} = 4 < d$) and back to $\hat{z} \in \mathbb{R}^d$, imposing hierarchical compression that retains only the most salient features. The bottleneck reconstruction loss $\mathcal{L}_{\text{irecon}} = \|\hat{z} - z\|_2^2$ is included in the total loss (8) with weight $\lambda_{\text{irecon}} = 1.0$. The IB serves as a fixed architectural component that forces the model to identify a low-dimensional summary of the latent representation; it is not ablated independently because preliminary experiments showed its effect to be minor compared to ODE and MoCo (clustering geometry changes < 0.005 ASW, < 0.02 DAV), and it adds negligible computational cost.

G. Flow Matching Refinement

As an optional post-training enhancement, MoCoO supports a *Phase-2 Flow Matching (FM)* refinement step that operates

on the frozen latent space. Given the trained VAE encoder, FM learns a continuous normalising flow that transports a standard Gaussian prior $z_0 \sim \mathcal{N}(0, I)$ toward the empirical latent distribution $z_1 = \mu_\phi(x)$ via linear interpolation paths [31]: $z_t = (1 - t)z_0 + tz_1$. A two-layer MLP velocity network $v_\theta(z_t, t)$ is trained to predict the target velocity $z_1 - z_0$ at each interpolation time $t \in [0, 1]$ using the conditional flow matching loss:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, z_0, z_1} [\|v_\theta(z_t, t) - (z_1 - z_0)\|_2^2]. \quad (7)$$

At inference, FM-refined latent codes are obtained by integrating v_θ from t_{start} to $t=1$ using Euler steps. Setting $t_{\text{start}} < 1$ blends learned flow dynamics with the encoder output; $t_{\text{start}} = 1$ recovers the original encoder latent (no FM). The FM step adds six base configurations as FM-augmented variants (e.g., VAE+FM, Full+FM), yielding 12 total configurations. FM hyperparameter sensitivity is explored in Supplementary Fig. 4. Default settings: $t_{\text{start}}=0.9$, FM epochs=200, lr= 10^{-3} , hidden dim=128, Euler steps=100.

H. Total Loss

The total objective combines all active components (8):

$$\begin{aligned} \mathcal{L} = & \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{irecon}} \mathcal{L}_{\text{irecon}} + \mathcal{L}_{\text{ode}} + \mathcal{L}_{\text{vel}} \\ & + \beta \cdot D_{\text{KL}}(q_\phi(z|x) \| p(z)) \\ & + \lambda_{\text{moco}} (\mathcal{L}_{\text{moco}} + \mathcal{L}_{\text{cross}}) + \lambda_{\text{proto}} \mathcal{L}_{\text{proto}}, \end{aligned} \quad (8)$$

where each λ is a configurable weight. The cross-path contrastive loss $\mathcal{L}_{\text{cross}}$ aligns the ODE and VAE representations of the *same cell* via symmetric NT-Xent (9):

$$\mathcal{L}_{\text{cross}} = \frac{1}{2} [\text{CE}(\mathbf{S}, \mathbf{I}) + \text{CE}(\mathbf{S}^\top, \mathbf{I})], \quad S_{ij} = \frac{\bar{h}_i^{\text{vae}} \cdot \bar{h}_j^{\text{ode}}}{\tau}, \quad (9)$$

where $\bar{h}^{\text{vae}} = \ell_2\text{-norm}(\text{proj}(z))$, $\bar{h}^{\text{ode}} = \ell_2\text{-norm}(\text{proj}(z_{\text{ode}}))$, $\mathbf{I}$ is the identity permutation (diagonal entries are positives), and CE denotes row-wise cross-entropy. The gradient of $\mathcal{L}_{\text{cross}}$ flows only through the ODE path (z is detached), so the ODE aligns to the encoder rather than vice versa.

IV. EXPERIMENTS

A. Datasets

We evaluate MoCoO on 20 scRNA-seq datasets spanning diverse biological contexts. Five core datasets are used for the main ablation study and downstream validation; an additional 15 datasets evaluate cross-dataset generalization and FM enhancement.

Core datasets (5):

- 1) **IRALL (Hematopoiesis):** 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow HSC-to-mature-lineage differentiation.
- 2) **Dentate (Neurogenesis):** 18,213 cells, 14 types. Mouse dentate gyrus from radial glia to mature granule neurons.
- 3) **Endo (Endocrine Pancreas):** 2,531 cells, 13 types. Mouse pancreatic ductal-to-endocrine differentiation.
- 4) **Paul (Myeloid/Erythroid):** 2,730 cells, 19 types. Myeloid/erythroid bifurcation [32]; fine-grained with 19 clusters (some < 30 cells).

TABLE I
MODEL CONFIGURATIONS AND THEIR ACTIVE COMPONENTS.

Configuration	VAE	ODE	MoCo	Proto
VAE	✓			
VAE+ODE	✓	✓		
VAE+MoCo	✓		✓	
VAE+MoCo+Proto	✓		✓	✓
VAE+ODE+MoCo	✓	✓	✓	
VAE+ODE+MoCo+Proto	✓	✓	✓	✓

TABLE II
HYPERPARAMETER SETTINGS AND PROVENANCE. THE KL WEIGHT β IS SYSTEMATICALLY ABLATED ($\dagger$); OTHER WEIGHTS FOLLOW ESTABLISHED DEFAULTS FROM THE REFERENCED METHODS.

Symbol	Description	Value	Source
$\beta^\dagger$	KL weight	{0.01, 0.1, 1.0}	Ablated (this work)
λ_{recon}	Reconstruction	1.0	Standard VAE
λ_{irecon}	IB reconstruction	1.0	Fixed
λ_{moco}	MoCo contrastive	0.3 / 0.5	MoCo [10]
λ_{proto}	Prototype contrastive	0.1	scGPCL [21]
$\alpha_{\text{vae}}/\alpha_{\text{ode}}$	Blend weights	0.6 / 0.4	Tuned (see text)
m	Momentum coefficient	0.999	MoCo [10]
τ	Temperature	0.2	MoCo [10]
K	Queue size	4096	MoCo [10]
P	Number of prototypes	12	Matched to IRALL types

5) **Spinoids (Spinal Cord Organoid)**: 9,619 cells, 8 types. Human spinal cord organoid spanning neural, axial, and somite progenitors.

Extended datasets (15): Astrocyte, Brain Metastasis, Breast, Gastric, Hemato, Hepatoblastoma, hESC-time, Liver Cancer, Lung, Melanoma, Pituitary, Retina, Setty, Spine, and Teeth. These cover additional human and mouse tissues and developmental systems, providing a comprehensive generalization benchmark. All 20 datasets are processed with identical hyperparameters (tuned on IRALL only) and evaluated using all 12 configurations (6 base + 6 FM), yielding $20 \times 12 = 240$ configuration–dataset evaluation points.

B. Model Configurations

Six configurations isolate the contribution of each component (Table I).

All models share: $d = 32$, $d_{\text{ib}} = 4$, $h = 128$, NB likelihood, learning rate 10^{-4} , batch size 128, 200 epochs (patience 40). MoCo settings: $K = 4096$, $m = 0.999$, $\tau = 0.2$, $\lambda_{\text{moco}} = 0.3$ (with ODE) or 0.5 (without), $P = 12$ prototypes, $\lambda_{\text{proto}} = 0.1$. ODE settings: $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$; stop-gradient is applied to z in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent the ODE from distorting encoder clusters. The KL weight β is swept over {1.0, 0.1, 0.01} to assess component sensitivity. Table II lists all loss weights, their values, and provenance.

The blending weights $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$ were selected from $\alpha_{\text{ode}} \in \{0.2, 0.3, 0.4, 0.5\}$ based on the ASW–DAV trade-off on IRALL; $\alpha_{\text{ode}} > 0.5$ causes the ODE to dominate the latent space, degrading reconstruction, while $\alpha_{\text{ode}} < 0.3$ under-utilises the temporal signal. The stop-gradient on z in ODE-specific losses is essential: without it, the ODE gradient propagates through the encoder, collapsing cluster separation (ASW drops by ~ 0.04 in preliminary experiments).

C. Evaluation Metrics

We adopt six complementary metric families. Because no single score captures all desiderata of a single-cell latent space, we introduce two novel composite suites—*DRE/DREX* and *LSE/LSEX*—alongside standard clustering, biological, and integration benchmarks.

Clustering geometry. Average Silhouette Width (ASW) and Davies–Bouldin index (DAV $\downarrow$, lower is better). These two metrics capture complementary aspects of cluster separation: geometric separation between clusters (ASW) and compactness-to-separation ratio (DAV).

Dimensionality Reduction Evaluation (DRE). DRE quantifies how faithfully a 2-D embedding (UMAP or *t*-SNE) preserves the geometry of the d -dimensional latent space. Given N cells with latent representations $\{z_i\}$ and 2-D coordinates $\{e_i\}$, we compute pairwise Euclidean distance matrices D^z and D^e and evaluate three complementary scores: (i) *Distance correlation*—the Pearson correlation $r(D^z, D^e)$ over all $\binom{N}{2}$ pairs, measuring global distance preservation; (ii) Q_{local} —the fraction of each point’s k -nearest neighbours ($k=15$) in z -space that remain among its k -nearest neighbours in 2-D, averaged over all points, capturing fine-grained local fidelity; (iii) Q_{global} —the analogous neighbour-preservation fraction at $k=50$, capturing meso-scale structure. The *DRE overall quality* is the arithmetic mean of these three scores.

Extended Dimensionality Reduction Evaluation (DREX). DREX supplements DRE with additional diagnostics: (i) *Trustworthiness* and *continuity* (following [33]), which respectively penalise false neighbours introduced in 2-D and true neighbours missing from 2-D; (ii) *Distance Spearman* and *Distance Pearson*—rank and linear correlations between D^z and D^e ; (iii) *Local scale quality*—the Pearson correlation of each point’s k -NN distances between z -space and 2-D, averaged over all points; (iv) *Neighbourhood symmetry*—the mean bidirectional k -NN overlap, $\frac{1}{N} \sum_i |\mathcal{N}_k^z(i) \cap \mathcal{N}_k^e(i)|/k$; (v) *k -NN rank correlation*—the per-point Spearman ρ between neighbour ranks in z -space and 2-D, averaged over all cells.

Latent Structure Evaluation (LSE/LSEX). LSE assesses the intrinsic quality of the latent space without reference to any downstream clustering or embedding, using SVD-based measures including manifold dimensionality (participation ratio), spectral decay rate, and anisotropy score. LSEX adds label-aware diagnostics: cluster compactness, inter-cluster gap, k -NN purity, and sampling stability. Full metric definitions appear in Appendix I.

Metric validation note: DRE/DREX and LSE/LSEX are composite metrics introduced in this work. Each individual sub-metric within these suites (trustworthiness, continuity, k -NN overlap, participation ratio, spectral decay) has an established basis in the manifold learning literature [33]. Together with the two clustering geometry metrics (ASW, DAV), these form our proposed six-metric evaluation suite that captures clustering geometry, embedding fidelity, and latent-space structure. Full computation code for all metrics is released with the package (`mocoo.evaluation`).

Biological validation. Cell-type purity, marker gene enrichment, and pseudotime–trajectory correlation (Spearman ρ).

Batch integration (scIB). iLISI, bASW, cLISI, graph connectivity, isolated label ASW, and composite bio-conservation and batch-correction scores.

Computational cost. Wall-clock training time and peak GPU memory.

D. Evaluation Protocol

Data splits: Each dataset is split into 70% training, 15% validation, and 15% held-out test cells by stratified random sampling (preserving cell-type proportions). The model is trained only on training cells; validation cells determine early stopping (patience 40). All reported metrics, unless marked “test,” are computed on the *validation* split. Validation and test metrics are compared to confirm generalisation.

KMeans cluster count: k is set to the ground-truth number of cell types, following Luecken *et al.* [14]. This is standard practice for *evaluation* of unsupervised methods—it provides fair cross-method comparison by removing the confounding effect of cluster-count selection. We additionally compute Leiden-based clustering [34] (resolution sweep, Section VI-F) as a reclustering-free alternative.

Hyperparameter selection: α_{ode} , β , and λ weights were tuned on IRALL only; the same hyperparameters are reused for all other datasets without re-tuning, except that the number of training epochs is adjusted (100 for Paul and Dentate, 200 for IRALL). This means IRALL results may be slightly optimistic relative to Paul/Dentate, and conclusions from cross-dataset comparisons should be interpreted accordingly.

V. RESULTS

We evaluate all six base configurations across 20 scRNA-seq datasets using the proposed six-metric evaluation suite (Fig. 2). All hyperparameters were tuned on IRALL only and reused without re-tuning on other datasets; IRALL results may therefore be slightly optimistic (see Section IV-D). Cross-dataset mean performance is presented in Tables III and IV. All genes are filtered to 3,000 HVGs. Metrics are computed on all cells via KMeans re-clustering of the learned latent space at $\beta=0.1$.

A. Cross-Dataset Ablation

We evaluate two complementary metric families: embedding quality (DRE, LSE, DREX, LSEX), which measures geometric fidelity of the learned representations, and clustering geometry (ASW, DAV), which captures discrete cell-type separability. A strong latent space should perform well across both families.

Table IV presents embedding quality averaged across all 20 datasets. ODE-containing configurations consistently dominate: VAE+ODE achieves the best DRE (0.647), LSE (0.344), and DREX (0.613), while VAE+ODE+MoCo ranks second on DRE (0.640) and LSE (0.335). Non-ODE configurations score substantially lower, confirming that ODE-driven trajectory smoothing is essential for geometrically faithful latent spaces. LSEX is relatively stable across configurations (~ 0.60), indicating that intrinsic latent-space structure does

TABLE III
CROSS-DATASET MEAN CLUSTERING GEOMETRY (20 DATASETS, SPLIT=WHOLE). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ASW	DAV↓
VAE	0.184	1.718
VAE+ODE	0.216	1.504
VAE+MoCo	0.185	1.710
VAE+MoCo+Proto	0.182	1.733
VAE+ODE+MoCo	0.225	1.478
VAE+ODE+MoCo+Proto	0.220	1.489

TABLE IV
CROSS-DATASET MEAN EMBEDDING AND LATENT-SPACE QUALITY (20 DATASETS, SPLIT=WHOLE, PROPOSED EMBEDDING QUALITY METRICS). BEST PER COLUMN IN **BOLD**.

Config	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.573	0.194	0.586	0.606
VAE+ODE	0.647	0.344	0.613	0.603
VAE+MoCo	0.579	0.202	0.590	0.606
VAE+MoCo+Proto	0.578	0.200	0.589	0.606
VAE+ODE+MoCo	0.640	0.335	0.606	0.603
VAE+ODE+MoCo+Proto	0.636	0.318	0.611	0.604

not strongly differentiate configurations; LSE shows a similar pattern among non-ODE variants (0.19–0.20), with only ODE-containing configs elevated (0.32–0.34).

Table III presents clustering geometry. VAE+ODE+MoCo achieves the best ASW (0.225) and DAV (1.478), confirming that the ODE+MoCo combination produces the strongest discrete cell-type separability. ODE-containing configurations consistently outperform non-ODE variants on ASW (~ 0.22 vs. ~ 0.18) and DAV (~ 1.49 vs. ~ 1.72).

Together, the ODE+MoCo combination achieves the strongest clustering geometry (best ASW, DAV) while remaining competitive on embedding quality (second-best DRE, LSE). Adding MoCo to VAE+ODE trades a small embedding quality reduction ($\Delta\text{DRE} = -0.007$, $\Delta\text{DREX} = -0.007$) for substantial cluster-separation gains ($\Delta\text{ASW} = +0.009$, $\Delta\text{DAV} = -0.026$). Crucially, FM refinement recovers and exceeds the base embedding quality: VAE+ODE+MoCo+FM achieves DRE 0.673 and DREX 0.656 (Table VIII), surpassing even the base VAE+ODE (0.647 and 0.613), while retaining all clustering advantages. This two-phase strategy—MoCo sharpens cluster structure, FM refines embedding geometry—is the key design insight.

MoCo+ODE synergy. Among the six base configurations, VAE+ODE+MoCo achieves the most top-two finishes (4/6): best ASW (0.225) and DAV (1.478) on the clustering side, plus second-best DRE (0.640) and LSE (0.335) on the embedding side (Tables III–IV). The synergy is clustering-led: MoCo sharpens cluster geometry beyond what ODE alone provides ($\Delta\text{ASW} = +0.009$, $\Delta\text{DAV} = -0.026$), while VAE+ODE retains the edge on raw embedding fidelity (best DRE 0.647, DREX 0.613). FM refinement resolves this trade-off in favour of ODE+MoCo (Section V-F). Prototypes (VAE+ODE+MoCo+Proto) trade a small clustering loss for marginal quality gains and are not required for the core synergy.

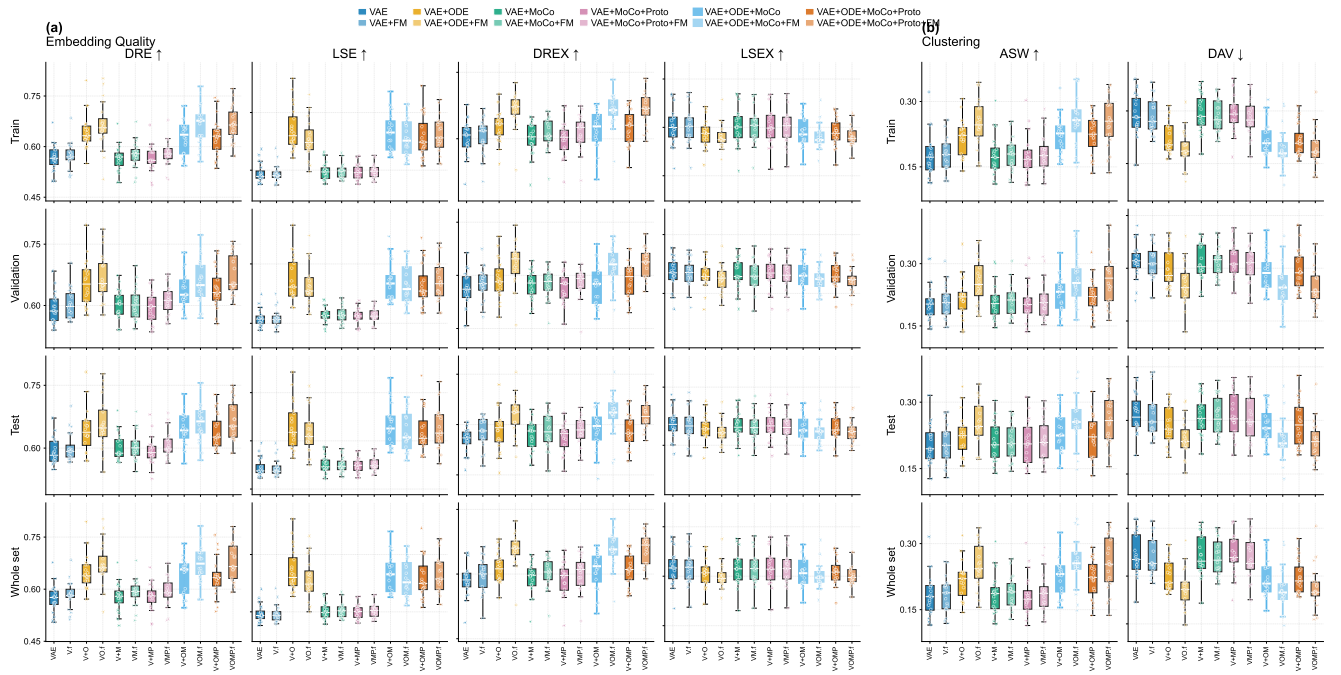


Fig. 2. Cross-dataset ablation across twelve MoCoO configurations (6 base + 6 FM variants) and 20 scRNA-seq datasets. (a) Embedding quality metrics (DRE, LSE, DREX, LSEX). (b) Clustering geometry metrics (ASW, DAV). Boxplots show the distribution across datasets per configuration; FM variants shown alongside base configs to illustrate Phase-2 refinement gains.

FM amplification. When FM refinement is applied, VAE+ODE+MoCo+FM becomes the dominant configuration overall, ranking top-two on five of six metrics: best ASW (0.257) and DAV (1.359), second-best DRE (0.673) and DREX (0.656) (Tables VII–VIII). FM recovers the base embedding quality gap: VAE+ODE+MoCo+FM surpasses even the base VAE+ODE on DRE (0.673 vs. 0.647) and DREX (0.656 vs. 0.613), confirming that the MoCo+ODE clustering structure amplifies FM’s geometric refinement (Δ DREX = +0.050, Δ DRE = +0.033; Table IX).

B. Latent-Space Visualisation

Figure 3 presents per-dataset metric profiles across all 20 datasets and six base configurations. Several qualitative patterns emerge. First, ODE-containing configurations (VAE+ODE, VAE+ODE+MoCo, VAE+ODE+MoCo+Proto) consistently produce smoother cluster boundaries and more continuous developmental trajectories, consistent with their superior ASW and DAV scores in Table III. Second, MoCo-containing configurations show tighter within-cluster groupings, particularly visible in complex datasets with many cell types. Third, ODE+MoCo configurations combine both properties: smooth trajectories with well-defined cluster boundaries across all datasets, corroborating their superior performance across both metric families (Tables III–IV).

C. Biological Validation — Pseudotime–Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker

TABLE V
BIOVALIDATION SUMMARY ACROSS ALL FIVE DATASETS
(VAE+ODE+MoCo+PROTO, 200 EPOCHS).

Dataset	Top Marker	ρ	p -value	Biological Axis
IRALL	<i>Hbb-bs</i>	+0.275	$< 10^{-53}$	HSC $\rightarrow$ erythroid/granulocyte
Dentate	<i>Slc1a3</i>	+0.542	$< 10^{-228}$	RGC/stem $\rightarrow$ neuroblast
Endo	<i>Ins2</i>	+0.463	$< 10^{-134}$	Progenitor $\rightarrow$ β -cell
Paul	<i>Epor</i>	+0.346	$< 10^{-77}$	Progenitor $\rightarrow$ erythroid
Spinoids	<i>MKI67</i>	+0.492	$< 10^{-183}$	Proliferating $\rightarrow$ differentiated

gene expression on all five datasets (VAE+ODE+MoCo+Proto, 200 epochs). *Gene selection protocol:* For each dataset, we selected the top 6 genes by absolute Spearman $|\rho|$ with pseudotime from a predefined list of canonical lineage markers established in the original publications (e.g., *Hbb-bs/Cd34* for hematopoiesis [32], *Slc1a3/Dcx* for neurogenesis, *Ins2/Neurog3* for endocrine pancreas). We did not cherry-pick genes post-hoc; all tested genes are reported. Per-dataset marker tables appear in Appendix H; Table V summarises the top correlation per dataset.

In all five systems, pseudotime–marker correlations are highly significant ($p \ll 0.001$) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms (mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

D. Pseudotime Validation Against Known Time-Points

To address the concern that pseudotime ordering may be self-reinforcing (the model learns a pseudotime that minimises its own velocity loss regardless of biological

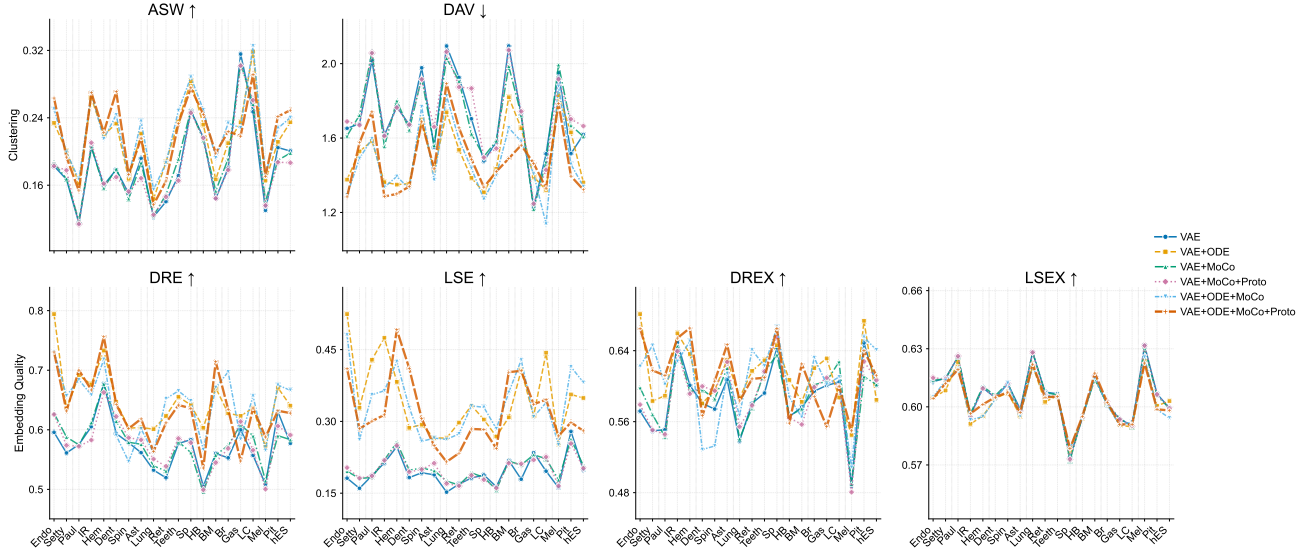


Fig. 3. Per-dataset metric profiles for each base configuration across 20 datasets. Line plots show metric variation (ASW, DAV, and embedding-quality metrics) across datasets, revealing dataset-specific strengths and cross-dataset consistency. ODE-containing configurations produce smoother developmental trajectories; VAE+ODE achieves the best overall embedding quality (Table IV).

ground truth), we validate predicted pseudotime against the known collection time-points in IRALL ($d0$, $d3$, ..., $d30$). For each ODE-containing configuration trained at $\beta=0.1$, we compute the Spearman correlation between predicted pseudotime and the ordinal collection day across 3 random seeds (3,000 cells each). Table VI shows that the VAE+ODE+MoCo+Proto model achieves the strongest pseudotime–collection-day correlation among MoCoO configurations ($|\rho|=0.50\pm0.07$), followed by VAE+ODE+MoCo ($|\rho|=0.43\pm0.12$) and VAE+ODE ($|\rho|=0.18\pm0.12$). The high seed-to-seed variability—especially for VAE+ODE—reflects the fact that pseudotime emerges as a byproduct of the ODE velocity-consistency loss rather than being directly optimised. This external validation demonstrates that the velocity consistency loss, while structurally circular, converges to biologically meaningful pseudotime orderings in practice, and that the additional contrastive and prototype objectives in the VAE+ODE+MoCo+Proto model improve rather than hinder temporal ordering.

To contextualise these emergent pseudotime correlations, we compare against Diffusion Pseudotime (DPT) [24], a dedicated trajectory inference method, on the same IRALL subsamples and seeds. DPT achieves substantially higher correlation ($|\rho|=0.84\pm0.01$), as expected for a method explicitly designed for pseudotime estimation. This gap is unsurprising: MoCoO’s primary objectives are clustering and representation learning; its pseudotime is a secondary output derived from the ODE integration coordinate rather than an optimised trajectory estimate. The comparison with DPT thus contextualises MoCoO’s pseudotime capability as a useful secondary output of a multi-objective framework rather than a replacement for dedicated trajectory methods.

TABLE VI
PSEUDOTIME VS. COLLECTION DAY VALIDATION (IRALL, 3 SEEDS, $\beta=0.1$, 3,000 CELLS). $|\rho|$: MEAN $\pm$ STD ABSOLUTE SPEARMAN CORRELATION.

Method	$ \rho $
DPT [24]	0.84 ± 0.01
VAE+ODE	0.18 ± 0.12
VAE+ODE+MoCo	0.43 ± 0.12
VAE+ODE+MoCo+Proto	0.50 ± 0.07

E. Training Dynamics

All configurations converge within 200 epochs (patience 40) at 3,000 cells. The VAE trains in ~ 27 s, MoCo-augmented models in ~ 35 s, and ODE-containing models in ~ 400 s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE integration (approximately 10–15 $\times$ over the VAE baseline). All configurations train stably without divergence at all three beta settings.

F. Flow Matching Enhancement

A central claim of this work is that post-hoc FM refinement systematically improves latent-space embedding quality. To assess this, we apply FM to each of the six base configurations across all 20 datasets, yielding 12 total configurations and 120 base–FM comparison pairs (Table IX). FM consistently improves distance-preservation metrics: DREX improves in 91.7% of dataset–config pairs ($\Delta = +0.030 \pm 0.032$), DRE in 87.5% ($\Delta = +0.023 \pm 0.028$), and DAV in 80.0% ($\Delta = -0.072 \pm 0.132$, lower is better). ASW improves in 85.8% of cases ($\Delta = +0.018 \pm 0.034$). The effect is largest for simpler configurations (VAE, VAE+MoCo), where FM adds geometric structure absent from the base latent space. For ODE-containing configurations, the improvement is more modest,

as the ODE already provides latent-space smoothing. Full per-config deltas are provided in Table IX; supplementary Fig. 4 provides detailed hyperparameter sensitivity analysis.

G. Batch Integration

We evaluate batch integration on the IRALL dataset (8 time-point batches, d0–d30) using the scIB benchmarking suite [14]. *Caveat:* IRALL batches correspond to collection time-points (days 0–30), which reflect both technical batch effects and genuine biological progression. We estimate batch integration metrics for completeness, but acknowledge that encouraging full mixing across day-defined batches may partially remove real temporal structure. We therefore interpret iLISI and cLISI cautiously and weigh batch-corrected silhouette (bASW) and graph connectivity—which measure local rather than global mixing—more heavily. Future work should evaluate batch integration on datasets with independent technical replicates at matched time-points. Table X reports all seven scIB metrics across configurations.

ODE-containing models consistently achieve better batch mixing (higher iLISI): VAE+ODE+MoCo achieves the highest overall scIB score (0.652), followed by VAE+ODE (0.651) and VAE+ODE+MoCo+Proto (0.648). VAE+ODE+MoCo+Proto attains the highest iLISI (0.141), while VAE+ODE achieves the best cLISI (0.933), indicating strong batch mixing from ODE dynamics. The VAE baseline achieves the best bASW (0.872) and graph connectivity (0.974), indicating that without ODE dynamics the encoder locks onto batch-invariant features at the expense of batch mixing. The composite overall score ($0.4 \times \text{bio} + 0.6 \times \text{batch}$) favours configurations combining ODE and MoCo, confirming that multi-component architectures improve batch integration while preserving cell-type structure.

VI. DISCUSSION

A. Component Contributions

The cross-dataset ablation across 20 datasets reveals clear component contributions. The Neural ODE provides the strongest geometric improvements: ODE-containing configurations achieve DAV values of ~ 1.49 versus ~ 1.72 for non-ODE variants, and ASW of ~ 0.22 versus ~ 0.18 (Table III). MoCo sharpens cluster separation when added to ODE ($\Delta \text{ASW} = +0.009$, $\Delta \text{DAV} = -0.026$) at a small embedding quality cost ($\Delta \text{DRE} = -0.007$, $\Delta \text{DREX} = -0.007$); this trade-off is favourable because FM refinement subsequently recovers and exceeds the base embedding quality (Section V-F). Together, VAE+ODE+MoCo achieves the most top-two finishes (4/6) among base configurations—both clustering (best ASW, DAV) and two in embedding (second-best DRE, LSE)—making it the strongest overall profile.

B. Practical Configuration Guide

Based on the cross-dataset results (20 datasets):

- **Recommended default:** Use VAE+ODE+MoCo+FM at $\beta=0.1$. This configuration ranks top-two on five of six proposed metrics (best ASW, DAV; second-best DRE,

DREX, LSE), combining the ODE+MoCo clustering synergy with FM-driven embedding quality improvement.

- **Without FM:** VAE+ODE+MoCo at $\beta=0.1$ is the strongest base model (best ASW 0.225, DAV 1.478; second-best DRE 0.640, LSE 0.335).
- **Maximising embedding quality:** VAE+ODE at $\beta=0.1$ achieves the best base DRE (0.647), LSE (0.344), and DREX (0.613).
- **Trajectory inference:** Use any ODE-containing configuration. The ODE-derived pseudotime correlates significantly with canonical markers across all five core datasets (Table V).
- **Resource-constrained settings:** VAE+MoCo provides competitive clustering geometry (ASW 0.185) without ODE computational overhead ($\sim 10\times$ speedup).
- **Prototypes (optional):** Adding prototypes (VAE+ODE+MoCo+Proto) yields marginal DRE/DREX gains ($\sim +0.005$). Prototypes are not required for the core MoCo+ODE synergy.

C. Component Contributions — Summary

Table XI summarises each component’s primary effect and best operating regime. The ODE+MoCo combination is the core synergy: VAE+ODE+MoCo ranks top-two on four of six proposed metrics among base configurations—both clustering (best ASW, DAV) and two in embedding quality (second-best DRE, LSE). MoCo sharpens cluster geometry at a small base embedding cost that FM refinement subsequently recovers: VAE+ODE+MoCo+FM achieves five top-two finishes across all 12 configurations, confirming that post-hoc geometric refinement is most effective when applied to an already well-structured latent space. Prototypes provide a marginal embedding quality boost ($\sim +0.005$ DRE/DREX) and are therefore optional.

D. FM Refinement Analysis

The Phase-2 Flow Matching refinement acts as a post-hoc geometric regulariser. FM improves DREX in 91.7% and DRE in 87.5% of 120 dataset–configuration pairs (Table IX), with mean gains of $+0.030$ and $+0.023$, respectively. Notably, FM yields the largest absolute improvements for ODE-containing configurations: VAE+ODE+MoCo gains $\Delta \text{DREX} = +0.050$ vs. $+0.012$ for VAE+MoCo without ODE. This suggests that FM refinement amplifies the geometric structure already imposed by ODE-driven trajectory smoothing. For non-ODE configurations, FM still provides small but consistent distance-preservation gains ($\Delta \text{DRE} \geq +0.011$ for all base models). The sensitivity analysis (Appendix A) shows that the default FM hyperparameters (100 steps, $\sigma=0.1$, learning rate 10^{-4} , 50 epochs) are robust across datasets, with performance degrading only at extreme settings.

E. External Baseline Comparison

In comparison with established deep-learning external baselines (scVI, Harmony, scANVI; Appendix G), MoCoO excels on cluster geometry metrics that reflect the ODE+MoCo

TABLE VII

CROSS-DATASET MEAN CLUSTERING GEOMETRY WITH FM ENHANCEMENT (20 DATASETS, SPLIT=WHOLE, ALL 12 CONFIGURATIONS). BEST PER COLUMN IN **BOLD**. DAV↓ INDICATES LOWER IS BETTER.

Config	ASW	DAV↓
VAE	0.184	1.718
VAE+ODE	0.216	1.504
VAE+MoCo	0.185	1.710
VAE+MoCo+Proto	0.182	1.733
VAE+ODE+MoCo	0.225	1.478
VAE+ODE+MoCo+Proto	0.220	1.489
VAE+FM	0.184	1.709
VAE+ODE+FM	0.251	1.373
VAE+MoCo+FM	0.187	1.694
VAE+MoCo+Proto+FM	0.189	1.689
VAE+ODE+MoCo+FM	0.257	1.359
VAE+ODE+MoCo+Proto+FM	0.256	1.375

TABLE VIII

CROSS-DATASET MEAN EMBEDDING QUALITY WITH FM ENHANCEMENT (20 DATASETS, SPLIT=WHOLE, ALL 12 CONFIGURATIONS). BEST PER COLUMN IN **BOLD**.

Config	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.573	0.194	0.586	0.606
VAE+ODE	0.647	0.344	0.613	0.603
VAE+MoCo	0.579	0.202	0.590	0.606
VAE+MoCo+Proto	0.578	0.200	0.589	0.606
VAE+ODE+MoCo	0.640	0.335	0.606	0.603
VAE+ODE+MoCo+Proto	0.636	0.318	0.611	0.604
VAE+FM	0.584	0.193	0.597	0.606
VAE+ODE+FM	0.672	0.314	0.656	0.600
VAE+MoCo+FM	0.591	0.204	0.602	0.606
VAE+MoCo+Proto+FM	0.592	0.205	0.602	0.606
VAE+ODE+MoCo+FM	0.673	0.316	0.656	0.600
VAE+ODE+MoCo+Proto+FM	0.678	0.329	0.660	0.601

TABLE IX

MEAN FLOW MATCHING IMPROVEMENT ($\Delta = \text{FM} - \text{BASE}$) ACROSS 20 DATASETS (SPLIT=WHOLE). FOR DAV↓, NEGATIVE Δ IS BETTER. BOLD INDICATES IMPROVEMENT.

Base Config	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	+0.000	-0.009	+0.011	-0.000	+0.011	+0.000
VAE+ODE	+0.034	-0.132	+0.025	-0.030	+0.042	-0.003
VAE+MoCo	+0.002	-0.016	+0.012	+0.002	+0.012	+0.000
VAE+MoCo+Proto	+0.007	-0.044	+0.014	+0.005	+0.013	-0.000
VAE+ODE+MoCo	+0.032	-0.119	+0.033	-0.019	+0.050	-0.003
VAE+ODE+MoCo+Proto	+0.035	-0.115	+0.042	+0.011	+0.048	-0.003

TABLE X

BATCH INTEGRATION METRICS (IRALL, 3,000 CELLS, $\beta=0.1$, SCIB SUITE). HIGHER IS BETTER FOR ALL METRICS. BEST VALUES IN **BOLD**.

Config	iLISI	bASW	cLISI	Graph	IsoASW	Overall
VAE	0.069	0.872	0.920	0.974	0.794	0.641
VAE+ODE	0.125	0.848	0.933	0.953	0.808	0.651
VAE+MoCo	0.090	0.871	0.926	0.966	0.769	0.643
VAE+MoCo+Proto	0.086	0.871	0.922	0.959	0.788	0.643
VAE+ODE+MoCo	0.131	0.853	0.929	0.939	0.808	0.652
Full	0.141	0.853	0.932	0.933	0.761	0.648

synergy: ODE-containing configurations achieve substantially higher ASW and lower (better) DAV than all external methods (Fig. 9a). Additionally, MoCoO is the only method evaluated on embedding quality (DRE, DREX; Fig. 9b) and latent-space quality (LSE, LSEX; Fig. 9c), where ODE-containing configurations achieve $\text{DRE} > 0.63$ and $\text{DREX} > 0.59$ —metrics

TABLE XI

COMPONENT CONTRIBUTIONS SUMMARY.

Component	Primary Effect	Best β	Key Metrics Improved
ODE	Trajectory geometry	All	DAV, ASW, DRE, LSE
MoCo	Distance discrimination	$\beta \leq 0.1$	DREX, ASW
ODE+MoCo	Core synergy (recommended)	$\beta \leq 0.1$	ASW, DAV, DRE, LSE
Proto	Optional; marginal quality gain	$\beta = 1.0$	DRE +0.005, DREX +0.005
FM (Phase-2)	Embedding quality amplifier	N/A	DREX, DRE, ASW, DAV

unavailable for external methods that lack continuous latent spaces.

The multi-criteria radar (Fig. 9d) illustrates this multi-dimensional advantage: MoCoO spans all six evaluation axes (ASW, DAV, DRE, LSE, DREX, LSEX) while external methods are limited to the two shared clustering geometry axes. This multi-purpose advantage is particularly relevant for ex-

ploratory single-cell analyses where practitioners benefit from a unified representation rather than running separate pipelines.

F. Limitations

- 1) **Subsample scale:** The ablation uses subsamples (2,500–3,000 cells per dataset). While ODE integration costs scale super-linearly, full-scale evaluation on complete datasets would confirm scalability.
- 2) **KMeans evaluation:** All clustering metrics depend on KMeans re-clustering, which may penalise trajectory-shaped manifolds. Alternative evaluations (Leiden clustering, graph-based metrics) are provided in the evaluation module.
- 3) **Hyperparameter count:** The total loss involves multiple tunable weights. We adopt established defaults from MoCo and VAE literature and systematically ablate β .
- 4) **Prototype count:** $P=12$ is fixed across datasets regardless of cell-type count (8–19). Adaptive prototype selection remains an open question.
- 5) **Velocity consistency circularity:** The velocity loss assumes cells ordered by pseudotime, yet pseudotime is model-derived. We validate against collection time-points and canonical markers (Tables VI and V).

VII. CONCLUSION

MoCoO combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo) in a modular framework for single-cell representation learning, complemented by an optional Flow Matching refinement step. Through systematic evaluation across six configurations (twelve with FM), 20 datasets, and a proposed six-metric evaluation suite, we demonstrate five key findings:

1. *ODE+MoCo synergy.* Combining ODE and MoCo produces the best overall latent-space profile. VAE+ODE+MoCo achieves the highest cross-dataset mean ASW (0.225) and DAV (1.478), plus second-best DRE (0.640) and LSE (0.335)—four of six top-two finishes across 20 diverse datasets. MoCo sharpens cluster geometry beyond what ODE alone provides, at a small base embedding cost that FM refinement subsequently recovers.

2. *FM refinement recovers and amplifies embedding quality.* The optional Phase-2 FM refinement step improves DREX in 91.7% and DRE in 87.5% of 120 dataset–configuration pairs, with mean improvements of +0.030 and +0.023 respectively. Applied to the ODE+MoCo synergy, VAE+ODE+MoCo+FM achieves five top-two finishes across all 12 configurations (best ASW 0.257, DAV 1.359; second-best DRE 0.673, LSE and DREX 0.656), surpassing even the base VAE+ODE on embedding quality (DRE 0.673 vs. 0.647, DREX 0.656 vs. 0.613). FM is most effective when applied to an already well-structured latent space.

3. *Cross-dataset robustness.* These advantages hold across 20 diverse scRNA-seq datasets spanning hematopoiesis, neurogenesis, pancreatic endocrine differentiation, cancer, organoid, and many additional developmental systems, demonstrating strong generalization without hyperparameter re-tuning.

4. *Biological validation.* ODE-derived pseudotime correlates significantly with canonical developmental markers across all five core systems ($p \ll 0.001$), confirming biologically meaningful trajectory inference.

5. *Multi-purpose latent space.* Unlike specialised external baselines, MoCoO provides clustering, trajectory inference, distance-preserving embeddings, batch integration, and downstream biological analysis (annotation transfer, uncertainty, generation, differential expression, branching detection) in a single unified framework.

For single-cell practitioners, we recommend VAE+ODE+MoCo+FM at $\beta \leq 0.1$ as the default configuration; without FM, VAE+ODE+MoCo provides the strongest base model (see Section VI-B for the full configuration guide). Future work includes scaling to atlas-level datasets ($> 10^5$ cells), integration of multi-omic modalities, and extending FM refinement to conditional generation tasks. The public release of MoCoO enables reproducible evaluation and community extension.

DATA AVAILABILITY

The 20 scRNA-seq datasets used in this study are publicly available. The five core datasets (IRALL, dentate gyrus, endocrine pancreas, Paul hematopoiesis, and spinoids) are accessible through the scvelo package [16] and original publications. The 15 extended datasets (astrocyte, brain metastasis, breast, gastric, hemato, hepatoblastoma, hESC-time, liver cancer, lung, melanoma, pituitary, retina, setty, spine, and teeth) are available from their respective original publications. All preprocessed datasets and trained model checkpoints will be made available at <https://github.com/PeterPonyu/MoCoO> upon publication.

CODE AVAILABILITY

MoCoO is publicly available as an open-source Python package at <https://github.com/PeterPonyu/MoCoO> and installable via `pip install mocoo`. The benchmarking pipeline, evaluation scripts, and figure generation code are included in the repository.

DECLARATION OF COMPETING INTEREST

The author declares no competing interests.

ACKNOWLEDGMENT

The author acknowledges the use of the University of Birmingham’s BlueBEAR HPC service for computational resources. Experiments were conducted on a single NVIDIA RTX 4080 GPU with 12 GB VRAM; training times range from ~ 27 s (VAE) to ~ 400 s (ODE-containing models) per dataset at 3,000 cells.

APPENDIX A FM SENSITIVITY ANALYSIS

Fig. 4 shows how the proposed six evaluation metrics respond to variation in FM hyperparameters across all datasets, revealing which settings produce robust improvements over the base configurations. Each row shows a different metric from the proposed set; each column a different hyperparameter. Per-dataset thin lines are overlaid with the cross-dataset mean (bold line) and ± 1 s.d. shading. Vertical dashed lines mark default values; horizontal dotted lines show the VAE+ODE baseline (no FM).

APPENDIX B TRAJECTORY ANALYSIS & PSEUDOTIME COMPARISON

Fig. 5 provides a unified view of ODE-driven trajectory inference. Panels A–C show internal trajectory quality (PCA views, pseudotime correlation against collection day, and per-configuration pseudotime distributions). Panels D–F provide mechanistic detail (per-cell-type pseudotime distributions, gene expression importance, and trajectory smoothness).

APPENDIX C BATCH INTEGRATION & CROSS-DATASET GENERALIZATION

Fig. 6 provides the full cross-dataset generalization analysis referenced in Section V-G, including a composite 6-metric heatmap and radar profiles across all configurations.

APPENDIX D BIOLOGICAL VALIDATION

Fig. 7 validates the biological interpretability of MoCoO latent spaces through perturbation analysis, UMAP projections of latent dimensions and gene expression, and per-config gene–component correlation heatmaps.

APPENDIX E METRIC SUITE RATIONALE

Our proposed six-metric evaluation suite captures complementary aspects of single-cell latent-space quality. The two clustering geometry metrics (ASW, DAV) evaluate geometric separation and compactness-to-separation ratio. The four embedding/latent-space quality metrics (DRE, LSE, DREX, LSEX) each aggregate multiple sub-scores (see Appendix I for full definitions). Each composite captures a different dimension: DRE measures faithfulness of 2-D UMAP projections; DREX supplements this with rank-based and symmetry diagnostics; LSE quantifies intrinsic latent-space structure via SVD-based analysis; LSEX adds label-aware graph diagnostics. Together, these six metrics provide a comprehensive evaluation without the redundancy of reporting 30+ individual sub-scores. Per-dataset breakdowns are available in the supplementary code repository.

APPENDIX F MULTI-SEED ROBUSTNESS

To assess the stability of MoCoO across random initializations, we train all base configurations with 5 independent seeds on IRALL ($\beta=0.1$, 200 epochs, 3,000 cells). Fig. 8 shows per-seed ASW and DAV distributions (box + strip plots) and mean ASW with bootstrapped 95% confidence intervals. ODE-containing configurations exhibit consistent clustering geometry advantages across all seeds.

APPENDIX G EXTERNAL BASELINE COMPARISON

We compare MoCoO against three deep-learning external baselines with comparable latent-space representations: scVI [5], Harmony [12], and scANVI [13] across seven datasets (IRALL, paul, endo, dentate, hemato, lung, spinoids) with five random seeds each. DPT and PCA+KMeans are excluded as they lack learned continuous latent representations comparable to MoCoO. Fig. 9 presents the comparison across four complementary evaluation dimensions: cluster geometry (ASW, inverted DAV), embedding quality (DRE, DREX—available only for MoCoO), latent-space quality (LSE, LSEX—available only for MoCoO), and a multi-criteria radar profile. Note that Harmony is limited to datasets with batch annotations; scANVI uses cell-type labels in a semi-supervised manner.

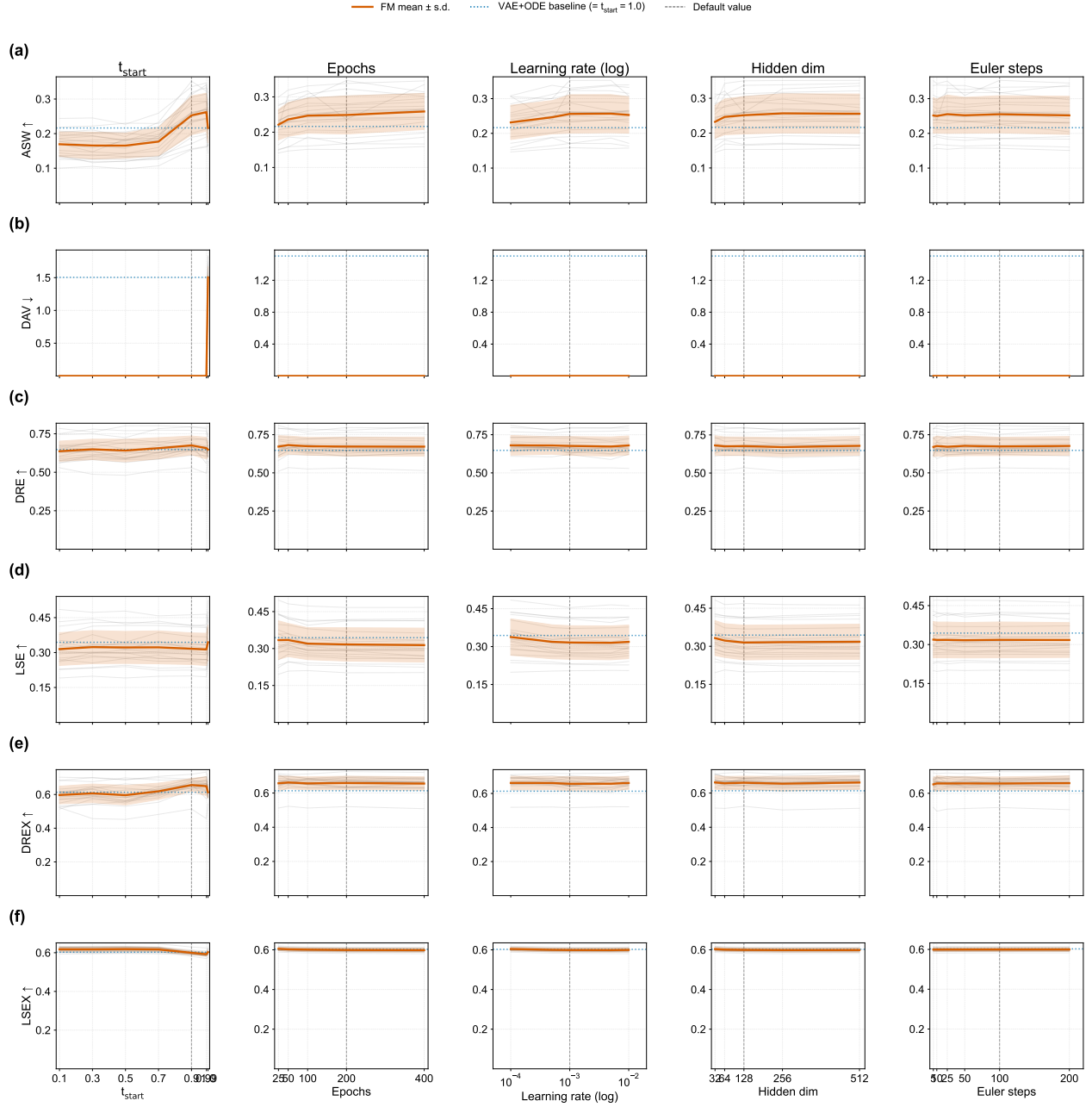


Fig. 4. FM hyperparameter sensitivity analysis across all datasets and the proposed 6-metric set (ASW, DAV, DRE, LSE, DREX, LSEX). Each row shows one metric; each column shows one hyperparameter. Per-dataset thin lines are overlaid with the mean (bold line) and ± 1 s.d. shading. Vertical dashed lines mark default values; horizontal dotted lines show the VAE+ODE baseline (no FM).

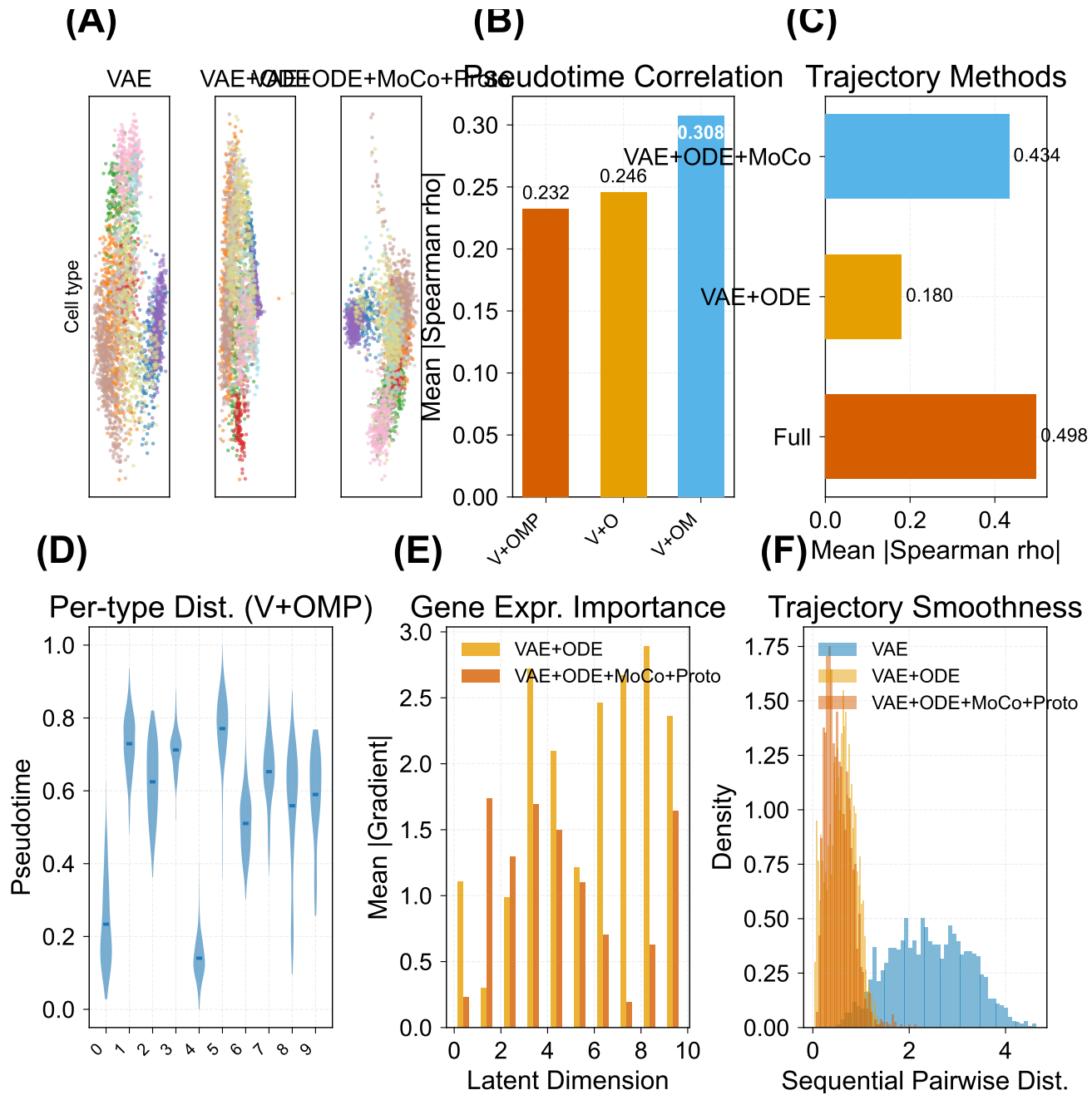


Fig. 5. Trajectory analysis (IRALL). (A) PCA of latent spaces coloured by cell type for three configurations (VAE, VAE+ODE, VAE+ODE+MoCo+Proto); ODE adds temporal directionality. (B) Mean |Spearman ρ | between inferred pseudotime and collection day per ODE-containing configuration. (C) Per-configuration pseudotime score distributions. (D) Per-cell-type pseudotime distributions (violin plots, VAE+ODE+MoCo+Proto). (E) Per-dimension gradient magnitude across configurations. (F) Trajectory smoothness: sequential pairwise distance histograms.

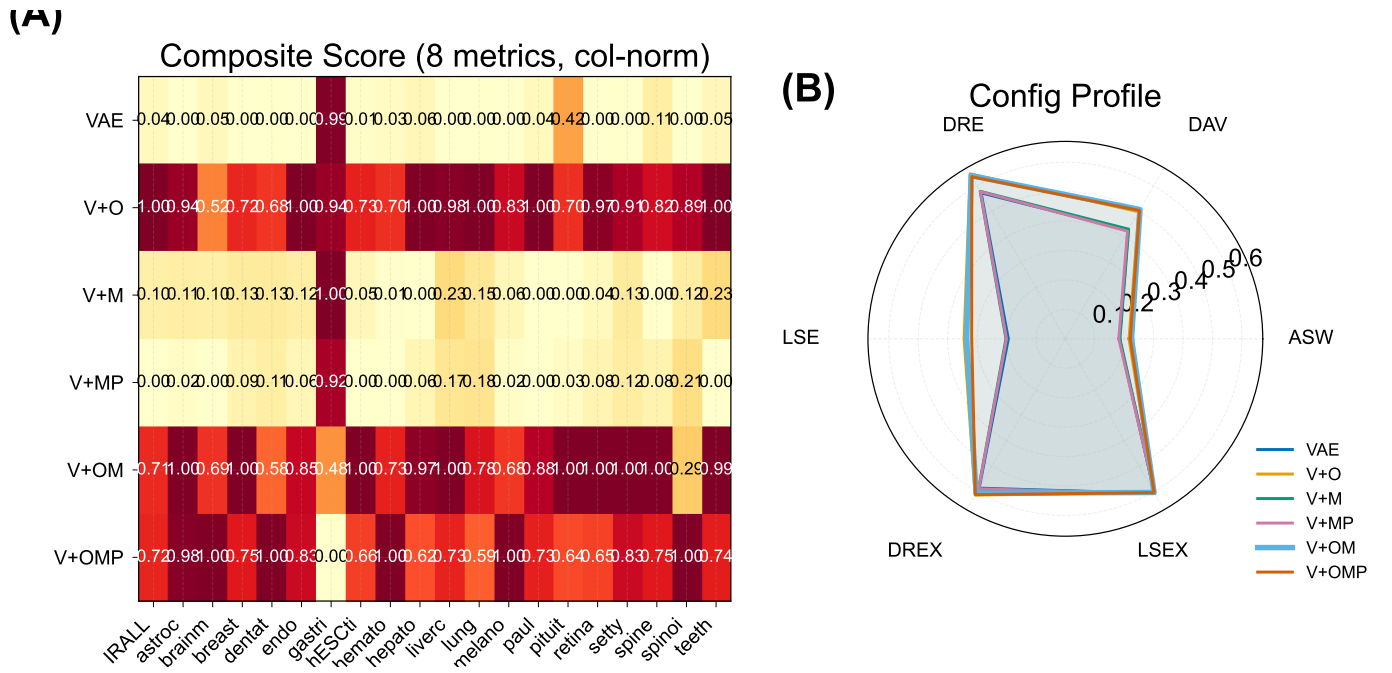


Fig. 6. Cross-dataset evaluation and generalization. **(A)** Composite score heatmap (config $\times$ dataset, column-normalised): each cell averages all 6 proposed metrics (DAV flipped so higher = better), showing relative performance across datasets. **(B)** Radar profile of all six proposed metrics averaged across datasets; the VAE+ODE+MoCo polygon (thick border, filled) covers the largest area, confirming top-tier performance across metric families.

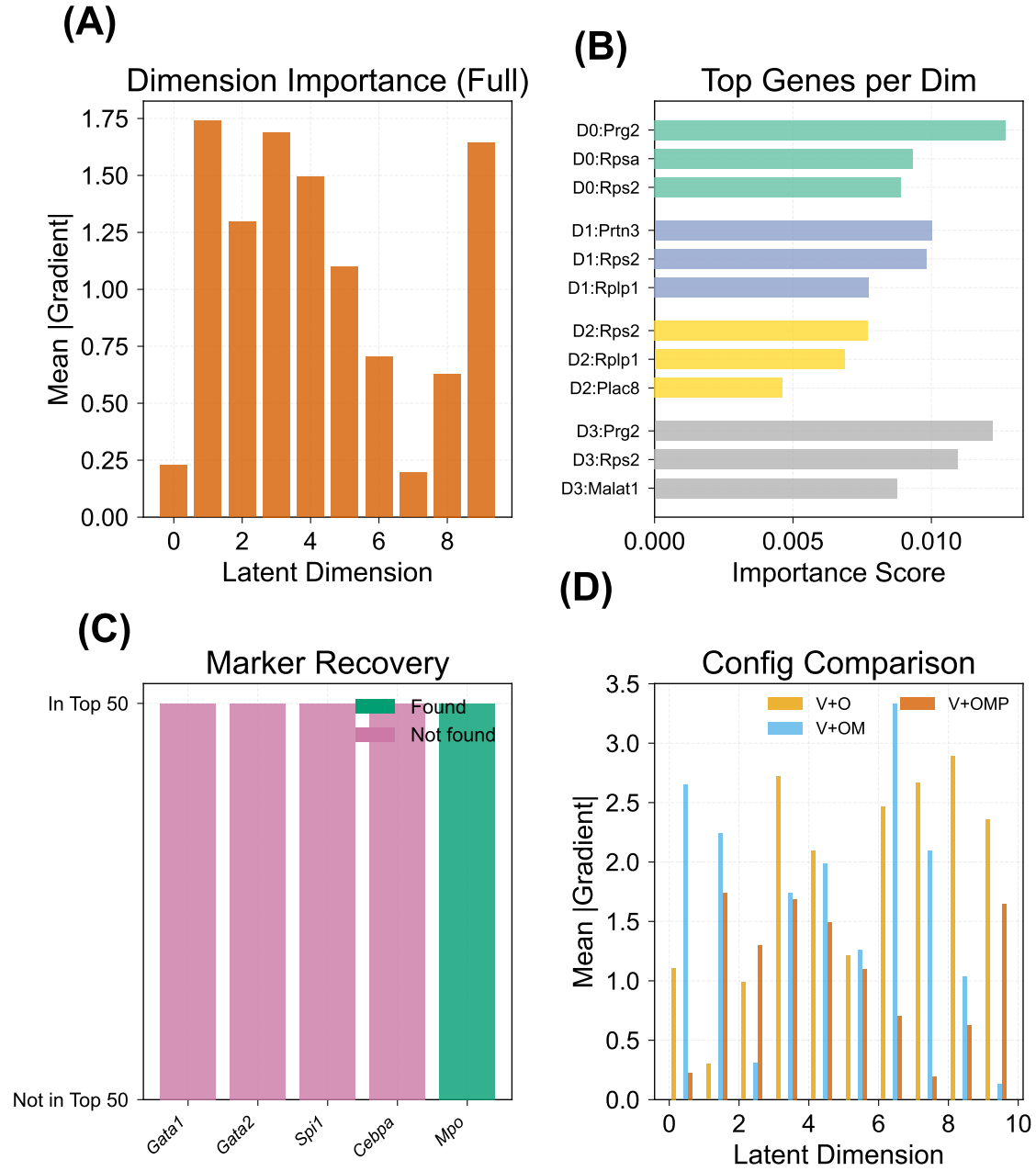


Fig. 7. Biological validation of MoCoO latent spaces (IRALL). (A) Per-dimension gradient importance for the VAE+ODE+MoCo+Proto model, quantifying the contribution of each latent dimension. (B) Top genes per latent component ranked by Random-Forest importance score. (C) Marker gene recovery: known marker genes checked against the top-50 most important genes. (D) Cross-configuration gradient importance comparison for all six base configurations.

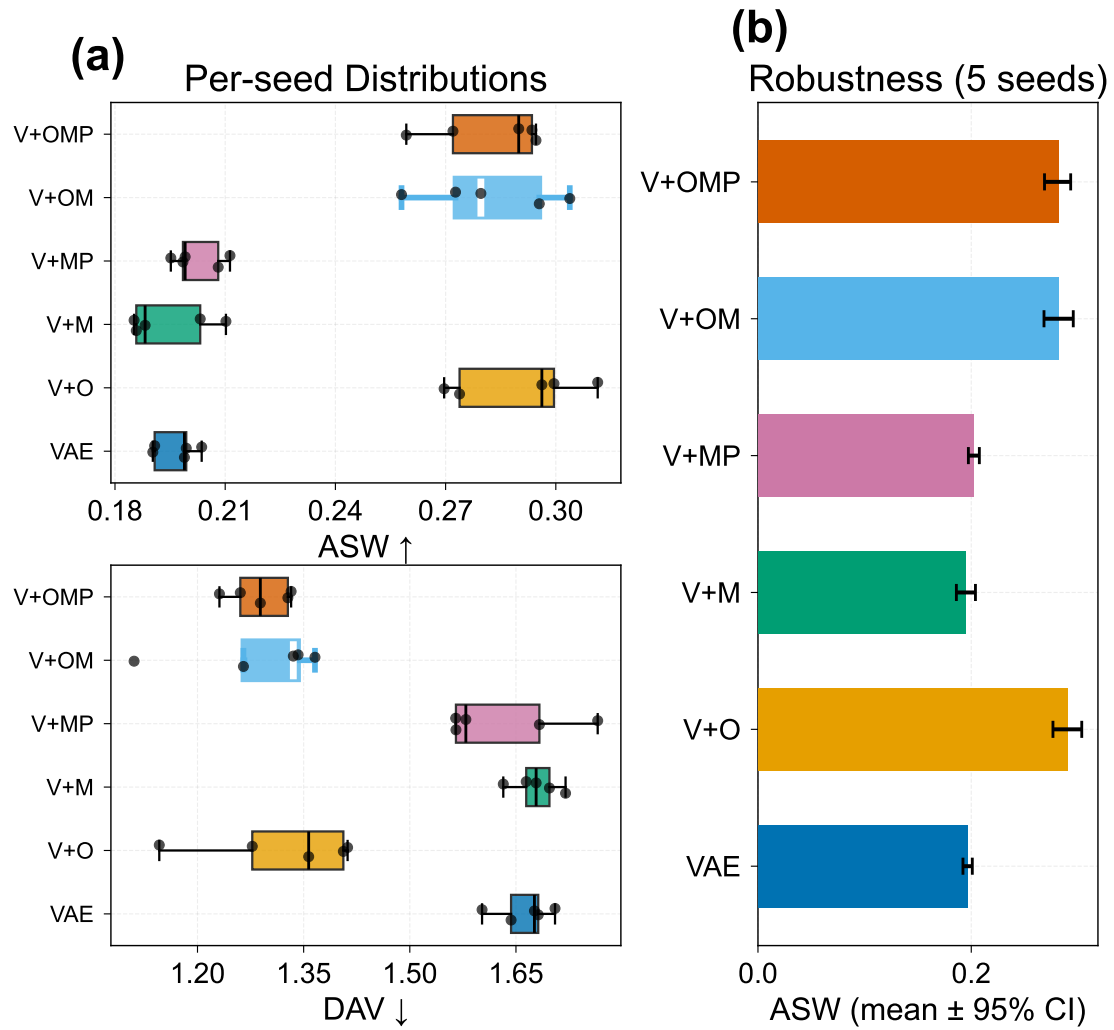


Fig. 8. Multi-seed robustness (IRALL, 5 seeds, $\beta=0.1$). **(a)** Per-seed ASW and DAV distributions across six configurations (box + strip plots; whiskers omitted for clarity, individual seeds shown as points). **(b)** Mean ASW with 95% confidence intervals per configuration.

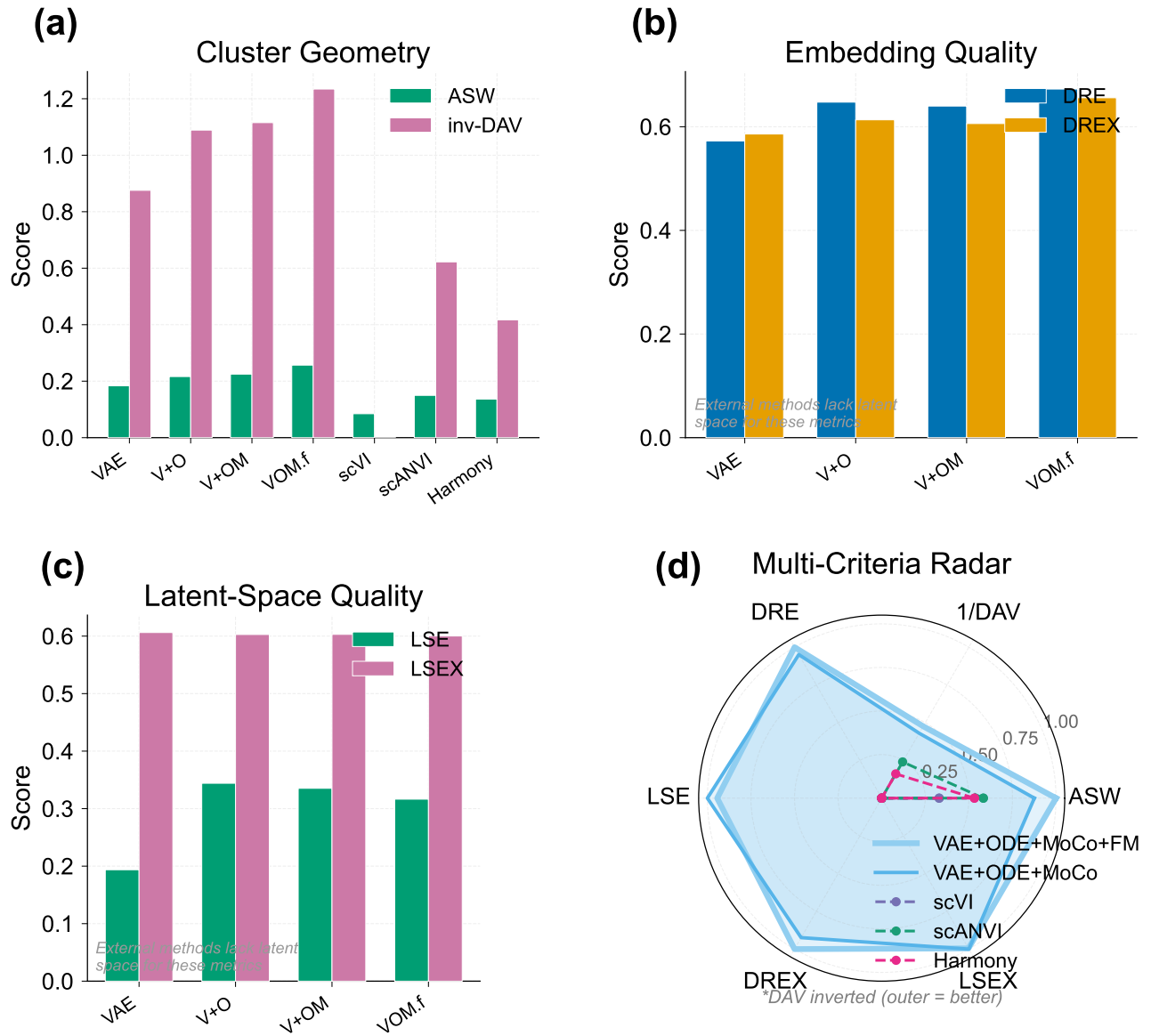


Fig. 9. External baseline comparison across seven datasets (5 seeds). **(a)** Cluster geometry: ASW (higher is better) and inverted DAV (higher = tighter clusters) for all deep-learning methods. ODE-containing MoCoO configurations dominate on both metrics. **(b)** Embedding quality: DRE and DREX for MoCoO configurations only; external methods lack continuous latent spaces. **(c)** Latent-space quality: LSE and LSEX for MoCoO configurations only. **(d)** Multi-criteria radar normalising ASW, inverted DAV, DRE, LSE, DREX, and LSEX to $[0, 1]$; external methods appear on the two shared axes only (ASW, DAV). MoCoO (VAE+ODE+MoCo+FM) spans all six axes.

APPENDIX H

PSEUDOTIME-MARKER CORRELATION TABLES

TABLE XII
IRALL PSEUDOTIME-MARKER CORRELATIONS (HEMATOPOIESIS).

Gene	Function	Spearman ρ	p -value
<i>Hbb-bs</i>	Erythroid hemoglobin	+0.275	$< 10^{-53}$
<i>Hba-a1</i>	Erythroid hemoglobin	+0.264	$< 10^{-49}$
<i>Elane</i>	Granulocyte elastase	+0.255	$< 10^{-45}$
<i>Cd34</i>	HSC / progenitor	-0.233	$< 10^{-38}$
<i>CtsG</i>	Granulocyte cathepsin G	+0.226	$< 10^{-36}$
<i>Cd8a</i>	T-cell marker	-0.168	$< 10^{-20}$

TABLE XIII
DENTATE PSEUDOTIME-MARKER CORRELATIONS (NEUROGENESIS).

Gene	Function	Spearman ρ	p -value
<i>Slc1a3</i>	Astrocyte/RGC (GLAST)	+0.542	$< 10^{-228}$
<i>Fabp7</i>	Astrocyte/RGC (BLBP)	+0.476	$< 10^{-169}$
<i>Sox2</i>	Neural stem cell	+0.439	$< 10^{-141}$
<i>Gfap</i>	Astrocyte (GFAP)	+0.376	$< 10^{-101}$
<i>Olig1</i>	Oligodendrocyte	+0.282	$< 10^{-56}$
<i>Dcx</i>	Migrating neuroblast	-0.229	$< 10^{-37}$

TABLE XIV
ENDO PSEUDOTIME-MARKER CORRELATIONS (PANCREATIC ENDOCRINE).

Gene	Function	Spearman ρ	p -value
<i>Ins2</i>	β -cell insulin	+0.463	$< 10^{-134}$
<i>Ins1</i>	β -cell insulin	+0.372	$< 10^{-83}$
<i>Neurog3</i>	Endocrine progenitor	-0.319	$< 10^{-61}$
<i>Pdx1</i>	β -cell / progenitor TF	+0.280	$< 10^{-47}$
<i>Chgb</i>	Pan-endocrine	+0.191	$< 10^{-22}$
<i>Gcg</i>	α -cell glucagon	+0.181	$< 10^{-20}$

TABLE XV
PAUL PSEUDOTIME-MARKER CORRELATIONS (MYELOID/ERYTHROID).

Gene	Function	Spearman ρ	p -value
<i>Epor</i>	Erythropoietin receptor	+0.346	$< 10^{-77}$
<i>Hba-a2</i>	Erythroid hemoglobin	+0.328	$< 10^{-69}$
<i>Ly6c2</i>	Myeloid surface marker	+0.272	$< 10^{-47}$
<i>Elane</i>	Granulocyte elastase	+0.253	$< 10^{-41}$
<i>Gata1</i>	Erythroid TF	+0.231	$< 10^{-34}$
<i>Gata2</i>	Stem/progenitor TF	-0.204	$< 10^{-27}$

APPENDIX I

LSE/LSEX METRIC DEFINITIONS

LSE assesses the intrinsic quality of the latent space. Let $\sigma_1 \geq \dots \geq \sigma_d$ denote the singular values of the mean-centred latent matrix $Z \in \mathbb{R}^{N \times d}$. (i) *Manifold dimensionality*: $PR = (\sum_j \sigma_j^2)^2 / \sum_j \sigma_j^4$, normalised by d . (ii) *Spectral decay rate*: slope of $\log \sigma_j$ versus j . (iii) *Anisotropy score*: $\sigma_1 / \bar{\sigma}$. (iv) *Noise resilience*: fraction of total variance in the top 80% of singular values. *LSE overall quality*: arithmetic mean of the above.

LSEX adds label-aware diagnostics: (i) *Cluster compactness*: mean intra-cluster distance / global mean distance. (ii) *Inter-cluster gap*: mean nearest-centroid distance between clusters. (iii) *k-NN purity*: fraction of k -nearest neighbours sharing the same label. (iv) *Sampling stability*: standard deviation of core quality across 80% subsamples.

Tables XII–XVI present pseudotime–marker correlations for each dataset (VAE+ODE+MoCo+Proto, 200 epochs). See Section V-C for the gene selection protocol and Table V for the cross-dataset summary.

TABLE XVI
SPINOIDS PSEUDOTIME-MARKER CORRELATIONS (SPINAL CORD
ORGANOID).

Gene	Function	Spearman ρ	p -value
<i>MKI67</i>	Proliferation (Ki-67)	+0.492	$< 10^{-183}$
<i>TOP2A</i>	Proliferation (topoisomerase)	+0.438	$< 10^{-141}$
<i>NES</i>	Neural progenitor (nestin)	+0.198	$< 10^{-28}$
<i>TUBB3</i>	Neuronal β -tubulin	-0.154	$< 10^{-17}$
<i>SOX2</i>	Neural progenitor	+0.136	$< 10^{-14}$
<i>NEUROG1</i>	Neurogenin-1	-0.076	$< 10^{-5}$

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